The second committee, headed by Kurup and with E V S Namboodiri as the member secretary, made haste to report that making a semi-cryogenic engine was easy. How wrong were they! It is indeed easier to handle the fuel (petroleum) for a semi-cryogenic than that of a cryogenic, which has liquid hydrogen as the fuel, but making a semi-cryogenic was not easy—at least in the late 1970s when the committee was formed.

When the committee recommended that the fuel for the 'easy' semi-cryogenic engine could be RP1, a petroleum fuel the Americans were using, I confronted them with a bit of history. A section of engineers in the US Army, I told them, was scared of handling unsymmetrical dimethylhydrazine (UDMH), which was then used in liquid propulsion engines, as the chemical was a known carcinogen and was not easy to store, transport or handle. So, they tried to shift to JP1, a petroleum-derived kerosene-like semi-cryogenic fuel (JP stands for NASA's Jet Propulsion Lab attached to California Institute of Technology).

The Americans found inconsistency in the performance of engines, with unclean combustion. Tests revealed that the behaviour of the fuel varied because it came from different oil wells. With each well, the sulphur and carbon content in the fuel varied. They refined the process several times, leading to products christened JP2, JP3 up to JP10. This final product called JP10 was renamed as RP1, the fuel Kurup and Namboodiri had now found to be the panacea for India's propulsion problems.

I asked them if we are able to get petroleum from the same well and keep the consistency intact; if we had the time for all that as PSLV was on the drawing board. There were no convincing answers. I advocated development of a full cryogenic engine first, then a semi-cryogenic. And that is the exact path ISRO took much later—not because I had suggested that, but because that's how the trial and error scientific method led us.

Brahm Prakash was convinced of my argument and gave the final nod to make the Vikas engine.

As Brahm Prakash was nearing retirement, the power struggle between Gowarikar and Muthunayagam intensified. Going by their academic background and experience, both were eligible for the post. But their history of playing politics was too well-known in the organisation, and that was a big negative when it came to vying for the top post at the country's premier space centre.